Fiscal Year:	FY 2012	**Task Last Updated:**	FY 07/16/2012
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events (ELFSTONE)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/20/2010	**End Date:**	09/19/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Douglas Matson is U.S. Team Member on ESA-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)" and "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)."		
COI Name (Institution):			
Grant/Contract No.:	NNX10AV27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with ESA-sponsored main projects as noted. Main project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany Subproject: Transformation kinetics - ISS Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the MSL-EML to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights on selected alloys. Long Duration microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS , Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	• Attended IWG-5 at ESA HQ in Noordwijk ND 12-14 December 2011 • Attended IWG-6 at DLR in Koln Germany 4-6 July 2012 Elected IWG Speaker (Chair) for ISS mission • Conducted ESL FeCo density tests at ESL-MSFC 9-13 January 2012 • Conducted ESL FeCo viscosity tests at ESL-MSFC 6-17 August 2012 • Graduated three MS students partially supported by the project. • July: hired Tufts undergrad to conduct glass-flux experiments on FeCo alloys to compare to ESL results (both at zero induced convection)
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Abstracts for Journals and Proceedings	Matson DM, Sander PJ. "Truncated dual-cap nucleation site development." TMS 2012 Meeting 141st Annual Meeting, Orlando, FL, March 11-15, 2012. Proceedings of the Materials Research in Microgravity Symposium. NASA Conference Publication. In press, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Lee J, Choufani P, Bradshaw RC, Hyers RW, Matson DM. "Containerless measurement of density and viscosity in Fe-Co alloys." TMS 2012 Meeting 141st Annual Meeting, Orlando, FL, March 11-15, 2012. Proceedings of the Materials Research in Microgravity Symposium. NASA Conference Publication. In press, 2012. , Jul-2012
Articles in Peer-reviewed Journals	Matson DM. "Items of Note from the Field, Profession, and Society. Materials science in microgravity." JOM Journal of the Minerals, Metals and Materials Society 2012 Jun;64(6):625. http://dx.doi.org/10.1007/s11837-012-0353-x , Jun-2012
Books/Book Chapters	Matson DM. "Nucleation in the Mushy Zone. Chapter 10." in "Solidification of Containerless Undercooled Melts." Ed. D.M. Herlach, D.M. Matson. Hoboken, N.J. : Wiley VCH, 2012. p. 213-238., Jul-2012
Books/Book Chapters	Hyers RW. "Computer-aided experiments in containerless processing of materials. Chapter 2." in "Solidification of Containerless Undercooled Melts." Ed. D.M. Herlach, D.M. Matson. Hoboken, N.J. : Wiley VCH, 2012. p. 31-50., Jul-2012

Books/Book Chapters	Matson DM. (Ed.). "Solidification of Containerless Undercooled Melts." Ed. D.M. Herlach, D.M. Matson. Hoboken, N.J. : Wiley VCH, 2012. 578 p. ISBN: 978-3-527-33122-2., Jul-2012
Dissertations and Theses	Fragoudaki R. (Roselita Fragoudakis). "Steel and GFRPs for beams in cyclic loading application." PhD thesis, Tufts University, February 2012. , Feb-2012